

[10 points]

Suppose that the splits at every level of quicksort are in the proportion $1 - s$ to s , where $0 < s < \frac{1}{2}$ is a constant. What is the minimum and maximum depth of a leaf in the recursion tree?

[6 points]

Write pseudocode for the procedure $\text{MIN-HEAPIFY}(A, i)$ and analyze the worst-case running time for this procedure.

[6 points]

Illustrate the operation of Radix-Sort on the following list of English words:

COW, DOG, SEA, RUG, ROW, MOW, BOX, TAB, BAR, EAR,
TAR, DIG, BIG, TEA, NOW, FOX

[8 points]

Use Master's Theorem to give tight asymptotic bounds for the following recurrences:

$$(a) \quad T(n) = 2T\left(\frac{n}{4}\right) + 1$$

$$(b) \quad T(n) = 2T\left(\frac{n}{4}\right) + \sqrt{n}$$

$$(c) \quad T(n) = 2T\left(\frac{n}{4}\right) + n$$

$$(d) \quad T(n) = 2T\left(\frac{n}{4}\right) + n^2$$